

Daniel Joseph Aronoff

Massachusetts Institute of Technology
Research Affiliate, Department of Economics

Curriculum Vitae, Academics
Updated: April 30, 2026
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Academic Appointments

2024– **Research Affiliate**, Department of Economics, MIT.
2022– Research Associate, The Laboratory for Economic Analysis and Design at MIT.
2022–2024 Research Scientist, MIT Media Lab.
2020– Research Collaborator, Digital Currency Initiative, MIT Media Lab.

Education

2022 **Massachusetts Institute of Technology**
 Ph.D. in Economics
 Advisors: Glenn Ellison, Robert Townsend and Nikhil Agarwal.
London School of Economics and Political Science
 B.Sc. in Philosophy and Economics (First Class Honors)
 Tutors: Peter Urbach (Philosophy) and John Sutton (Economics).
University of Michigan
 Attended
 Studied Russian Intellectual History
 Advisor: Arthur Mendel.

Research Interests

Financial market structure; digital currencies; cryptography; environmental market design.

Publications

Journal Articles

Aronoff, D. and W. Rafey (May 2026). Conservation Priorities and Environmental Offsets: Markets for Florida Wetlands. *American Economic Review*. [Link](#) [PDF](#).

Books and Chapters

Aronoff, D. and I. Ardis (2024). ADESS: A Proof-of-Work Protocol to Deter Double-Spend Attacks. *Advances in Information and Communication*. Ed. by K. Arai. Vol. 920. Lecture Notes in Networks and Systems. FICC 2024 Berlin. Cham: Springer, pp. 131–157. [Link](#) [PDF](#).
 Aronoff, D. (2016a). *A Theory of Accumulation and Secular Stagnation*. New York, NY: Palgrave Pivot. [Link](#).
 — (2016b). *The Financial Crisis Reconsidered: The Mercantilist Origin of Secular Stagnation and Boom-Bust Cycles*. New York, NY: Palgrave Macmillan New York. [Link](#).

Conference Proceedings

Aronoff, D., A. Bhat, P. Chatzigiannis, M. Minaei, S. Raghuraman, R. M. Townsend, and N. X.-Y. Zhang (2025b). SoK: Fully-homomorphic encryption in smart contracts. Conference Poster. FHE.org2025 Sofia, Bulgaria. [Link](#).
 Aronoff, D. (2024a). ADESS: A Proof-of-Work Protocol to Deter Double-Spend Attacks. Conference slides. FICC2024 Berlin. [Link](#).

Industry Publications

- Aronoff, D. J., F. Calabia, A. Brownworth, A. Samuel, and N. Narula (2026). The Hidden Plumbing of Stablecoins: Financial and Technological Risks in the GENIUS Act Era. MIT Media Lab Digital Currency Initiative research project sponsored by MITRE. [Link](#).
- Aronoff, D. (2023). *DeFi for the Securities Industry*. Finadium Research Report. [Link](#).
- Aronoff, D. and J. Galper (2022). Bank Stablecoins vs. Central Bank Digital Currencies for Capital Markets. Finadium Research Report. [Link](#).

Working Papers / Preprints

- Aronoff, D. and M. Virza (2026). A Synthetic Repo Contract. MIT working paper. [Link](#).
- Aronoff, D. (2025a). Mechanism Design for Hashrate Externalities: Implementing Optimal Security in Proof-of-Work. unpublished manuscript MIT. [Link](#).
- Aronoff, D., A. Bhat, P. Chatzigiannis, M. Minaei, S. Raghuraman, R. M. Townsend, and N. X.-Y. Zhang (2025a). SoK: Fully-homomorphic encryption in smart contracts. Cryptology ePrint Archive, Paper 2025/527. Accepted paper: FC26 Decentralized Finance (DeFi) Workshop, in association with Financial Cryptography. [Link](#).
- Aronoff, D. and R. M. Townsend (2025). A Smart-Contract to Resolve Multiple Equilibrium in an Intermediated Repo Trade. arXiv preprint. [Link](#).
- Aronoff, D., R. M. Townsend, and M. Virza (2025). TradeMech: Multilateral Netting Without Altering Counterparty Risk. unpublished manuscript, MIT. [Link](#).
- Aronoff, D. J., R. M. Townsend, and M. Virza (2025). *RepoMech: A Method to Reduce the Balance-Sheet Impact of Repo Intermediation*. [Link](#).
- Aronoff, D. (2024d). Targeted Nakamoto: A Bitcoin Protocol to Balance Network Security and Carbon Emissions. arXiv preprint, version 4. [Link](#).
- Moroz, D. J., D. J. Aronoff, N. Narula, and D. C. Parkes (2020). Double-Spend Counterattacks: Threat of Retaliation in Proof-of-Work Systems. arXiv preprint. [Link](#).

Thesis

- Aronoff, D. (2022). “Essays on Incentive Designs to Improve Market Performance”. PhD thesis; available from MIT DSpace. PhD thesis. Cambridge, MA, USA: Massachusetts Institute of Technology. [Link](#).

Recorded Talks and Video Presentations

- Daniel Aronoff and Madars Virza (Oct. 2025). *AI-Driven, Post-Quantum Data Privacy Research Spotlight*. Presentation at MIT-ILP conference Securing the Future: AI-Driven Post-Quantum Data Privacy. [Link](#).
- Aronoff, D. (June 2024b). *Incorporating Cryptographic Methods in Economics*. Presentation MIT-ILP NextGen Security Conference. [Link](#).
- Aronoff, D., J. Flynn, and E. Golding (Nov. 2023). *R&D Fintech Innovation*. Panel Discussion at 2023 MIT Research and Development Conference. [Link](#).
- Aronoff, D. (Sept. 2021b). *Rethinking Currency with Daniel Aronoff*. Invited Lecture: Rethinking Economics NL. [Link](#).

Slide Presentations

- Aronoff, D. (Oct. 2024c). SCBX Cryptography Use Cases: Credit Score Averaging with Differential Privacy and FHE. Presentation to SCBX R&D Department. [Link](#).
- (2021a). A Causal Inference Algorithm for Heterogeneous Treatment Effects. Presentation to MIT Econometrics Lunch. [Link](#).

Blog Posts

- Aronoff, D., A. Brownworth, J. Jin, A. Samuel, and N. Narula (July 2025). *Stablecoins and the Limits of Existing Analogies*. MIT Media Lab Digital Currency Initiative. [Link](#).

Aronoff, D., A. Brownworth, A. Samuel, and N. Narula (July 2025). *The GENIUS Act is Now Law. What's Missing?* MIT Media Lab Digital Currency Initiative. [Link](#).

Aronoff, D., A. Samuel, and N. Narula (July 2025a). *1:1 Redemptions for Some, Not All*. MIT Media Lab Digital Currency Initiative. [Link](#).

— (July 2025b). *Will Stablecoins Impact the US Treasury Market?* MIT Media Lab Digital Currency Initiative. [Link](#).

Project Sites

Aronoff, D. (July 2025b). *Targeted Nakamoto: Project site*. MIT Media Lab Digital Currency Initiative. [Link](#).

Aronoff, D. and I. Ardis (July 2025). *ADESS: Project site*. MIT Media Lab Digital Currency Initiative. [Link](#).

Professional Service

Committees	2022: Advances in Financial Technologies (AFT2022) Program Committee
Refereeing	2020 - 2023: <i>Cryptoeconomic Systems</i> .
Academic Boards	2003 - 2008: Grand Valley State University Grand Rapids, MI. Board of Trustees. 2006 – 2008: Vice-Chair. 2007 - 2010: Roeper School Bloomfield Hills, MI. Board of Trustees

Consulting

2024– Client: SCB^X. Project: An AI-Driven Post-Quantum Cryptographically Secure Workflow for Collaborative Credit Scoring.

References

Available upon request.